

Image-Identifiable Genomic Subspaces: A Linear–Gaussian Model of Radiogenomic Recoverability

Radiogenomic Information-Theoretic Bounds Project

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Abstract

We develop a linear–Gaussian generative model linking patient-level genomic profiles $g \in \mathbb{R}^p$ to imaging phenotypes $x \in \mathbb{R}^d$, and use it to ask a sharper question than “does imaging predict gene j ?” Instead we ask *which low-dimensional genomic subspaces are image-identifiable*. Under the model every quantity of interest is available in closed form: the Bayes-optimal estimator $\mathbb{E}[g \mid x]$, the posterior covariance $\Sigma_{G \mid X}$, the mutual information $I(g; x)$, and a per-direction *recoverability score* $R(v) \in [0, 1]$. We show that the directions of maximal recoverability solve a generalized eigenproblem that coincides exactly with canonical correlation analysis (CCA) between the two modalities, and that the recoverability of the i -th canonical genomic direction equals the squared canonical correlation ρ_i^2 . We then address estimation in the regime $p \gg n$ via regularized covariances, reduced-rank regression, and the probabilistic-CCA latent-variable form, and describe cross-validated and permutation-based procedures for honest inference. The companion notebook `notebooks/radiogenomic_recoverability.ipynb` implements the full pipeline on a pair of `AnnData` objects, with a synthetic-data fallback that has known ground truth.

1 Setup and notation

We observe n patients. For patient i let $g_i \in \mathbb{R}^p$ be a vector of genomic features (e.g. expression, somatic mutation burden, copy-number scores) and $x_i \in \mathbb{R}^d$ a vector of imaging phenotypes (radiomic descriptors, deep features, or structured reads). Stack the observations into data matrices $\mathbf{G} \in \mathbb{R}^{n \times p}$ and $\mathbf{X} \in \mathbb{R}^{n \times d}$, with row i equal to $g_i^\top$ and $x_i^\top$ respectively. Random variables are written in uppercase (G, X), realizations in lowercase (g, x). Throughout we assume both modalities have been centered, so all means are zero; recentering is handled by the estimators of Section 10.

We use $\Sigma_G \in \mathbb{R}^{p \times p}$, $\Sigma_X \in \mathbb{R}^{d \times d}$ for the marginal covariances, $\Sigma_{GX} = \text{Cov}(G, X) \in \mathbb{R}^{p \times d}$ for the cross-covariance, and $\Sigma_{XG} = \Sigma_{GX}^\top$. For a symmetric positive semidefinite S we write $S^{1/2}$ for its symmetric square root and $S^{-1/2}$ for the inverse thereof.

2 The linear–Gaussian generative model

The genomic state is drawn from a Gaussian prior and the imaging phenotype is a noisy linear readout of it:

$$G \sim \mathcal{N}(0, \Sigma_G), \tag{1}$$

$$X \mid G \sim \mathcal{N}(AG, \sigma^2 I_d), \quad A \in \mathbb{R}^{d \times p}. \tag{2}$$

The matrix A encodes how genomic variation is expressed in the image; the isotropic term $\sigma^2 I_d$ collects everything in the image that genomics does not drive — biological variation outside the measured panel, acquisition and reconstruction noise, and feature-extraction error. Equivalently,

$$X = AG + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d), \quad \varepsilon \perp G. \quad (3)$$

Equations (1)–(3) are deliberately the simplest model with a non-trivial answer. Their value is that *every* downstream quantity is closed-form, so the model functions as an exact upper bound: no estimator — linear or nonlinear — can recover genomics from imaging better than the posterior derived below (under these assumptions). Section 10 relaxes the isotropic-noise and known-parameter assumptions for practical fitting.

3 Marginal and joint laws

Proposition 1 (Joint Gaussian law). *Under (1)–(3),*

$$\begin{pmatrix} G \\ X \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \Sigma_G & \Sigma_G A^\top \\ A \Sigma_G & A \Sigma_G A^\top + \sigma^2 I_d \end{pmatrix}\right). \quad (4)$$

In particular,

$$X \sim \mathcal{N}(0, \Sigma_X), \quad \Sigma_X = A \Sigma_G A^\top + \sigma^2 I_d, \quad (5)$$

and the cross-covariance is

$$\Sigma_{GX} = \text{Cov}(G, X) = \Sigma_G A^\top. \quad (6)$$

Proof. Since

$$X = AG + \varepsilon,$$

where

$$G \sim \mathcal{N}(0, \Sigma_G), \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d),$$

and G and ε are independent, the stacked vector

$$\begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} G \\ AG + \varepsilon \end{pmatrix}$$

is jointly Gaussian, as it is an affine transformation of independent Gaussian variables.

It therefore suffices to compute its mean and covariance. Since both G and ε are centered,

$$\mathbb{E}[G] = 0, \quad \mathbb{E}[\varepsilon] = 0,$$

we obtain

$$\mathbb{E}[X] = A\mathbb{E}[G] + \mathbb{E}[\varepsilon] = 0,$$

and hence

$$\mathbb{E} \begin{pmatrix} G \\ X \end{pmatrix} = 0.$$

Next, the covariance matrix has block form

$$\text{Cov} \begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} \text{Cov}(G, G) & \text{Cov}(G, X) \\ \text{Cov}(X, G) & \text{Cov}(X, X) \end{pmatrix}.$$

The upper-left block is simply

$$\text{Cov}(G, G) = \Sigma_G.$$

For the upper-right block, using $X = AG + \varepsilon$,

$$\text{Cov}(G, X) = \text{Cov}(G, AG + \varepsilon).$$

By linearity of covariance,

$$\text{Cov}(G, X) = \text{Cov}(G, AG) + \text{Cov}(G, \varepsilon).$$

Since G and ε are independent,

$$\text{Cov}(G, \varepsilon) = 0,$$

and therefore

$$\text{Cov}(G, X) = \text{Cov}(G, AG).$$

Using the identity

$$\text{Cov}(U, BV) = \text{Cov}(U, V)B^\top,$$

we obtain

$$\text{Cov}(G, AG) = \Sigma_G A^\top.$$

Hence

$$\Sigma_{GX} = \text{Cov}(G, X) = \Sigma_G A^\top.$$

The lower-left block is the transpose:

$$\text{Cov}(X, G) = \text{Cov}(G, X)^\top = A \Sigma_G.$$

Finally, for the lower-right block,

$$\text{Cov}(X, X) = \text{Cov}(AG + \varepsilon).$$

Expanding the covariance,

$$\text{Cov}(X, X) = \text{Cov}(AG) + \text{Cov}(\varepsilon),$$

since the cross terms vanish by independence. Using

$$\text{Cov}(AG) = A \Sigma_G A^\top,$$

together with

$$\text{Cov}(\varepsilon) = \sigma^2 I_d,$$

gives

$$\text{Cov}(X, X) = A \Sigma_G A^\top + \sigma^2 I_d.$$

Therefore

$$\Sigma_X = A \Sigma_G A^\top + \sigma^2 I_d.$$

Collecting the covariance blocks yields

$$\text{Cov} \begin{pmatrix} G \\ X \end{pmatrix} = \begin{pmatrix} \Sigma_G & \Sigma_G A^\top \\ A \Sigma_G & A \Sigma_G A^\top + \sigma^2 I_d \end{pmatrix},$$

which proves the claim. $\square$

Equation (5) is the model's first interpretive statement: imaging covariance decomposes additively into a *genomics-driven* component $A \Sigma_G A^\top$ and an *irreducible* component $\sigma^2 I_d$. The cross-covariance $\Sigma_{GX} = \Sigma_G A^\top$ is the central estimable object — it names which genomic directions co-vary with imaging at all.

4 Bayes-optimal recovery of genomics from imaging

Proposition 2 (Posterior). *The posterior of genomics given imaging is Gaussian, $G \mid X \sim \mathcal{N}(\hat{g}(X), \Sigma_{G|X})$, with*

$$\hat{g}(X) = \mathbb{E}[G \mid X] = \Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I_d)^{-1} X = BX, \quad (7)$$

$$\Sigma_{G|X} = \Sigma_G - \Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I_d)^{-1} A \Sigma_G. \quad (8)$$

Equivalently, in precision form,

$$\Sigma_{G|X}^{-1} = \Sigma_G^{-1} + \sigma^{-2} A^\top A, \quad B = \sigma^{-2} \Sigma_{G|X} A^\top. \quad (9)$$

Proof. Equations (7)–(8) are the standard Gaussian conditioning formulas applied to Proposition 1. The precision form follows from the Woodbury identity: $(\Sigma_G^{-1} + \sigma^{-2} A^\top A)^{-1} = \Sigma_G - \Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I_d)^{-1} A \Sigma_G$, and then $B = \Sigma_G A^\top \Sigma_X^{-1} = \sigma^{-2} \Sigma_{G|X} A^\top$ by the push-through identity $\Sigma_G A^\top (A \Sigma_G A^\top + \sigma^2 I)^{-1} = (\Sigma_G^{-1} + \sigma^{-2} A^\top A)^{-1} \sigma^{-2} A^\top$. $\square$

The estimator $\hat{g}(X) = BX$ is the Bayes-optimal predictor under squared-error loss; because the model is jointly Gaussian it is also the optimal predictor over *all* measurable functions of X , not merely linear ones. Thus $\Sigma_{G|X}$ is a hard floor on residual genomic uncertainty: **no algorithm can do better**. The precision form (9) makes the accounting transparent — imaging *adds* information $\sigma^{-2} A^\top A$ to the prior precision Σ_G^{-1} .

5 Mutual information and the SNR decomposition

Proposition 3 (Mutual information). *The mutual information between the two modalities is*

$$I(G; X) = \frac{1}{2} \log \frac{\det \Sigma_G}{\det \Sigma_{G|X}} = \frac{1}{2} \log \det (I_d + \sigma^{-2} A \Sigma_G A^\top). \quad (10)$$

Let $\lambda_1 \geq \dots \geq \lambda_r > 0$ be the nonzero eigenvalues of $A \Sigma_G A^\top$. Then

$$I(G; X) = \frac{1}{2} \sum_{i=1}^r \log \left(1 + \frac{\lambda_i}{\sigma^2} \right). \quad (11)$$

Proof. For jointly Gaussian (G, X) , $I(G; X) = \frac{1}{2} \log (\det \Sigma_G / \det \Sigma_{G|X})$. Substituting (9), $\det \Sigma_{G|X} = \det \Sigma_G / \det (I_p + \sigma^{-2} \Sigma_G^{1/2} A^\top A \Sigma_G^{1/2})$, and Sylvester’s determinant identity moves A to the other side to give $\det (I_d + \sigma^{-2} A \Sigma_G A^\top)$. Diagonalizing $A \Sigma_G A^\top$ yields (11). $\square$

Equation (11) is the cleanest reading of the model: each image-visible genomic direction contributes $\frac{1}{2} \log(1 + \text{SNR}_i)$ bits (nats), where $\text{SNR}_i = \lambda_i / \sigma^2$ is the signal-to-noise ratio of that direction. Directions whose genomic signal is small relative to imaging noise contribute essentially nothing, regardless of how many genes load on them. Figure 1 illustrates the decomposition (11) on the synthetic instance introduced below: the per-direction bit contribution falls off rapidly after the first handful of canonical components, and the cumulative total saturates at $I(G; X) \approx 6.45$ bits out of the analytic upper bound implied by the model.

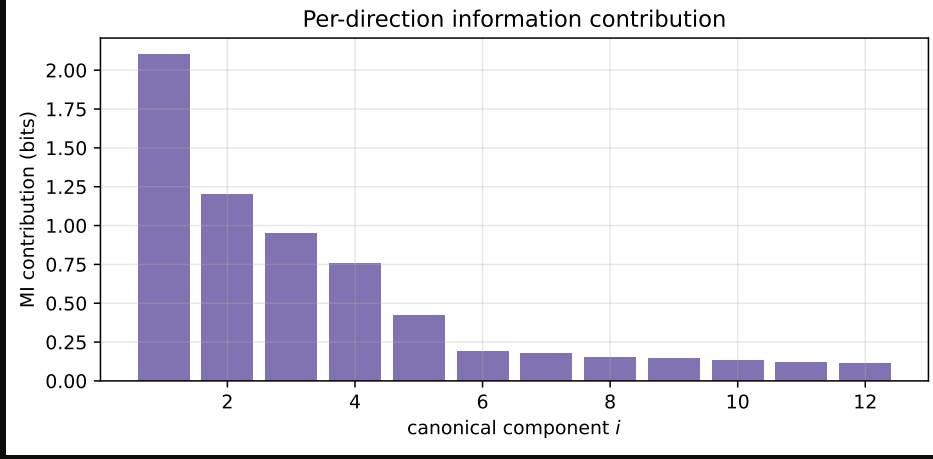


Figure 1: Per-direction contribution to $I(G; X)$ on a synthetic linear-Gaussian instance ($n = 800$, $p = 80$, $d = 40$). Each bar is $\frac{1}{2} \log(1 + d_i^2)$ in bits; the dashed line is the cumulative total.

6 Canonical coordinates and the recoverability spectrum

The eigenvalues λ_i are most interpretable after whitening. Define the whitened genomic variable $U = \Sigma_G^{-1/2} G \sim \mathcal{N}(0, I_p)$ and the *whitened transfer operator*

$$M := \sigma^{-1} A \Sigma_G^{1/2} \in \mathbb{R}^{d \times p}, \quad M = \sum_i d_i u_i v_i^\top \quad (\text{SVD}), \quad (12)$$

with left/right singular vectors $u_i \in \mathbb{R}^d$, $v_i \in \mathbb{R}^p$ and singular values $d_1 \geq d_2 \geq \dots \geq 0$.

Proposition 4 (Canonical decomposition of recoverability). *With M as in (12), define the genomic canonical direction $a_i := \Sigma_G^{-1/2} v_i$ and the canonical score $S_i := a_i^\top G = v_i^\top U$. Then:*

- (i) S_i has prior variance 1 and posterior variance $\text{Var}(S_i | X) = (1 + d_i^2)^{-1}$;
- (ii) the canonical correlation between S_i and imaging is $\rho_i = d_i / \sqrt{1 + d_i^2}$, equivalently $d_i^2 = \rho_i^2 / (1 - \rho_i^2)$;
- (iii) the eigenvalues of Section 5 satisfy $\lambda_i / \sigma^2 = d_i^2$, so

$$I(G; X) = \frac{1}{2} \sum_i \log(1 + d_i^2) = -\frac{1}{2} \sum_i \log(1 - \rho_i^2). \quad (13)$$

Proof. In whitened coordinates the likelihood is $X | U \sim \mathcal{N}(\sigma M U, \sigma^2 I_d)$, since $AG = A \Sigma_G^{1/2} U = \sigma M U$. The posterior precision of U is $I_p + \sigma^{-2} (\sigma M)^\top (\sigma M) = I_p + M^\top M = V(I_p + D^2) V^\top$, so $\text{Var}(U | X) = V(I_p + D^2)^{-1} V^\top$ and $\text{Var}(S_i | X) = v_i^\top \text{Var}(U | X) v_i = (1 + d_i^2)^{-1}$, giving (i). For (ii), $\text{Cov}(U, X) = \mathbb{E}[U(\sigma M U + \varepsilon)^\top] = \sigma M^\top$, so $\text{Cov}(S_i, X) = \sigma v_i^\top M^\top = \sigma d_i u_i^\top$. The matched imaging variate $T_i := u_i^\top X$ has $\text{Var}(T_i) = u_i^\top \Sigma_X u_i = \sigma^2 (1 + d_i^2)$ (using $\Sigma_X = \sigma^2 (M M^\top + I_d)$) and $\text{Cov}(S_i, T_i) = \sigma d_i$, hence $\rho_i = \text{corr}(S_i, T_i) = d_i / \sqrt{1 + d_i^2}$. Part (iii) is immediate from $A \Sigma_G A^\top = \sigma^2 M M^\top$ and $\det(I_d + M M^\top) = \prod_i (1 + d_i^2)$. $\square$

Part (i) gives the headline identity. Define the recoverability of a genomic score as one minus the fraction of its variance that survives observing imaging (formalized in Section 7); then

$$R(a_i) = 1 - \text{Var}(S_i \mid X) = 1 - \frac{1}{1 + d_i^2} = \frac{d_i^2}{1 + d_i^2} = \rho_i^2. \quad (14)$$

The recoverability of the i -th canonical genomic direction is exactly the squared canonical correlation. This is what reduces the whole analysis to CCA between $\mathbf{G}$ and $\mathbf{X}$ (Section 10), and it means A and σ^2 never have to be estimated individually — only the three second moments $\Sigma_G, \Sigma_X, \Sigma_{GX}$ enter.

7 Recoverability of a named genomic direction

In practice one often cares about a specific, biologically named genomic score $Y = v^\top G$ for a fixed $v \in \mathbb{R}^p$ — a pathway signature, a polygenic score, a single gene's expression ($v = e_j$).

Definition 1 (Recoverability score). *For $v \in \mathbb{R}^p$ with $v^\top \Sigma_G v > 0$,*

$$R(v) := 1 - \frac{v^\top \Sigma_{G|X} v}{v^\top \Sigma_G v} = \frac{v^\top (\Sigma_G - \Sigma_{G|X}) v}{v^\top \Sigma_G v} \in [0, 1]. \quad (15)$$

$R(v) = 0$ means imaging carries no information about $Y = v^\top G$; $R(v) = 1$ means imaging determines it exactly. The score is scale-invariant in v and, by the Gaussian conditional-variance identity, equals the population R^2 of the Bayes-optimal predictor of Y from X : $R(v) = 1 - \mathbb{E}[\text{Var}(Y \mid X)] / \text{Var}(Y)$. The matrix at the core of (15) has the convenient moment-only form

$$\Sigma_G - \Sigma_{G|X} = \Sigma_G A^\top \Sigma_X^{-1} A \Sigma_G = \Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}. \quad (16)$$

8 The recoverability eigenproblem

The natural project-level question — *which* genomic directions does imaging see best — is the variational problem

$$\max_{v \neq 0} R(v) = \max_{v \neq 0} \frac{v^\top (\Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG}) v}{v^\top \Sigma_G v}. \quad (17)$$

Proposition 5 (Recoverability eigenproblem = CCA). *The stationary points of (17) are the solutions of the generalized eigenproblem*

$$\Sigma_{GX} \Sigma_X^{-1} \Sigma_{XG} v = \mu \Sigma_G v. \quad (18)$$

Its eigenvalues are $\mu_i = \rho_i^2 = d_i^2 / (1 + d_i^2)$ and its eigenvectors are the canonical genomic directions a_i of Proposition 4, Σ_G -orthonormal ($a_i^\top \Sigma_G a_j = \delta_{ij}$). Thus the ordered solutions of (18) are precisely the canonical correlation directions between G and X , and $\max_v R(v) = \rho_1^2$.

Proof. (18) is the first-order condition of the Rayleigh quotient (17). Substitute $v = \Sigma_G^{-1/2} w$ and $\Sigma_{GX} = \Sigma_G A^\top$; the generalized eigenproblem becomes the ordinary one $\Sigma_G^{1/2} A^\top \Sigma_X^{-1} A \Sigma_G^{1/2} w = \mu w$. Now $A \Sigma_G^{1/2} = \sigma M$ and $\Sigma_X = \sigma^2 (M M^\top + I_d)$, so

$$\Sigma_G^{1/2} A^\top \Sigma_X^{-1} A \Sigma_G^{1/2} = \sigma M^\top \cdot \sigma^{-2} (M M^\top + I_d)^{-1} \cdot \sigma M = M^\top (M M^\top + I_d)^{-1} M.$$

With the SVD $M = UDV^\top$ this equals $V D^2(D^2 + I)^{-1} V^\top$, whose eigenvalues are $d_i^2/(1 + d_i^2) = \rho_i^2$ with eigenvectors v_i . Hence $w = v_i$ and $v = \Sigma_G^{-1/2} v_i = a_i$; the Σ_G -orthonormality $a_i^\top \Sigma_G a_j = v_i^\top v_j = \delta_{ij}$ is immediate. $\square$

So the deliverable of the analysis is a *recoverability spectrum* $\rho_1^2 \geq \rho_2^2 \geq \dots$ together with the genomic directions a_i that achieve it. Reading the loadings of $a_1, a_2, \dots$ back onto genes or pathways answers “which biology is image-identifiable”: the leading directions typically align with coarse, spatially expressed programs — proliferation, hypoxia, angiogenesis, necrosis, immune infiltration, tumor burden, stromal remodeling — rather than with individual transcripts.

Illustration on synthetic data. Figure 2 shows the recoverability spectrum on a synthetic radiogenomic instance generated from the linear–Gaussian model of Section 2 ($n = 800$, $p = 80$, $d = 40$, latent rank $k = 5$ with prescribed $\{\rho_i^*\}$). The estimated $\hat{\rho}_i^2$ closely match the planted values for the identifiable directions and decay toward the high-dimensional bulk thereafter, validating Proposition 5. Figure 3 reads the same estimator against the analytic ground truth derived from Propositions 2–4: the recovered posterior mean $\hat{B}X$, posterior-variance retention, and direction alignment all track the closed-form predictions.

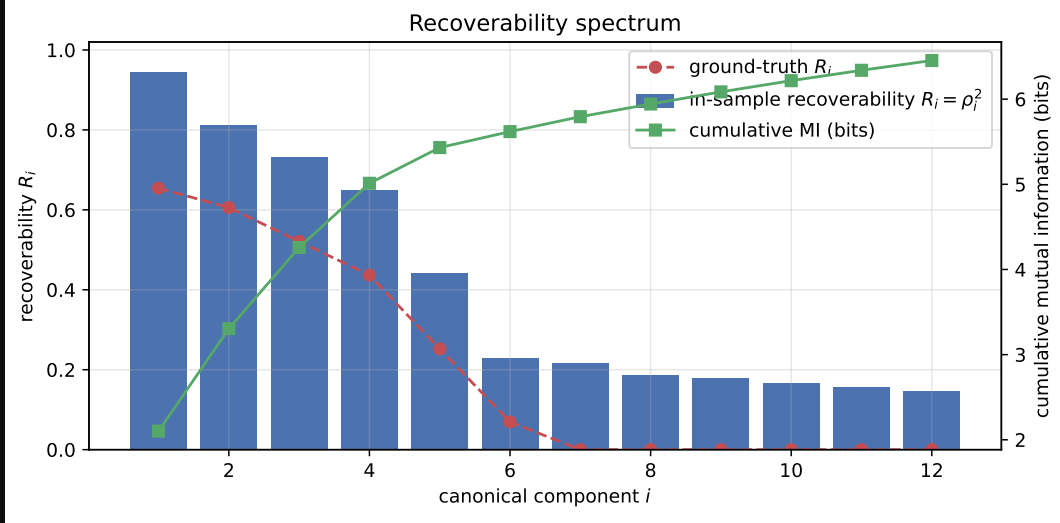


Figure 2: Recoverability spectrum on a synthetic linear–Gaussian instance ($n = 800$, $p = 80$, $d = 40$). The leading $\hat{\rho}_i^2$ recovers the planted canonical structure; later components fall to the high-dimensional bulk.

9 Rank limits and identifiability

Proposition 6 (Rank limit). *The recoverable signal matrix has rank*

$$\text{rank}(\Sigma_G - \Sigma_{G|X}) = \text{rank}(A\Sigma_G A^\top) \leq \min(d, p, \text{rank } A, \text{rank } \Sigma_G). \quad (19)$$

Consequently at most $\min(d, p, \text{rank } A, \text{rank } \Sigma_G)$ canonical correlations are nonzero, and the image-identifiable genomic subspace $\text{span}\{a_i : \rho_i > 0\}$ has dimension bounded by the same quantity.

This is the conceptual core. Even with $p = 20,000$ measured genes, imaging can expose only a projection of genomics whose dimension is capped by the rank of the transfer map A — if a CT

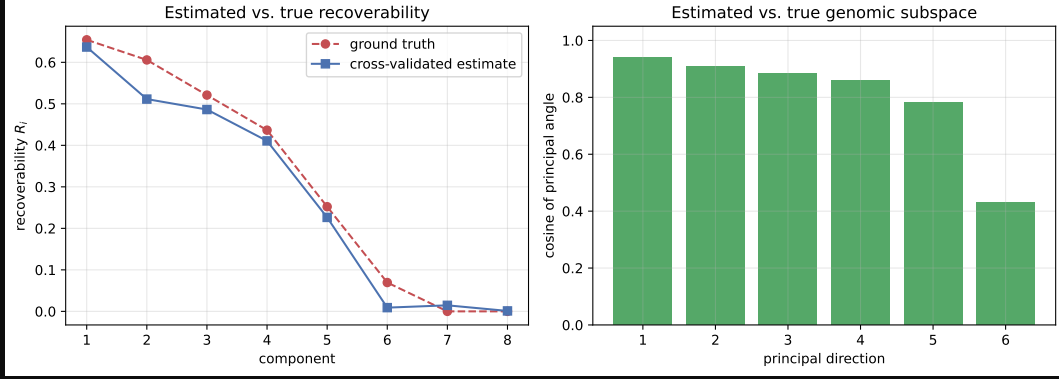


Figure 3: Empirical-vs-analytic check on the same synthetic instance. Estimated canonical correlations, recoverability per direction, and imaging variance attributable to genomics all match the closed-form quantities from Propositions 2–4.

phenotype reflects, say, 20 macroscopic biological axes, then no model recovers 20,000 independent molecular variables from it. The recoverable dimension is a property of the *biology–imaging channel*, not of the estimator. A data-processing remark: imaging features are themselves a deterministic, lossy function $x = \phi(\text{image})$; any such post-processing can only shrink $\text{rank}(\Sigma_{GX})$ and lower each ρ_i , never raise them.

10 Estimation from finite samples

In applications p and d may rival or exceed n , the noise is not isotropic, and A, Σ_G, σ^2 are unknown. Because Propositions 4 and 5 require only $\Sigma_G, \Sigma_X, \Sigma_{GX}$, we estimate those directly.

10.1 Regularized covariances

Replace sample covariances by shrinkage estimators (e.g. Ledoit–Wolf), $\hat{\Sigma}_G = (1 - \alpha_G) S_G + \alpha_G \nu_G I_p$ and likewise $\hat{\Sigma}_X$, with $S_G = n^{-1} \mathbf{G}^\top \mathbf{G}$ the centered sample covariance; the cross-covariance is $\hat{\Sigma}_{GX} = n^{-1} \mathbf{G}^\top \mathbf{X}$. Shrinkage keeps $\hat{\Sigma}_G, \hat{\Sigma}_X$ invertible and well-conditioned, which is required for the whitening in (12). When p is very large it is practical to first project genomics onto its top principal components (or onto curated pathway scores), reducing p to a manageable $p' \ll n$ before estimation; the rank limit of Proposition 6 means little is lost provided p' exceeds the recoverable dimension.

10.2 Reduced-rank and ridge regression

The transfer map, if needed explicitly, is the multivariate regression of X on G . Ridge gives $\hat{A}^\top = (\mathbf{G}^\top \mathbf{G} + \gamma I_p)^{-1} \mathbf{G}^\top \mathbf{X}$, and $\hat{\sigma}^2$ follows from the residual covariance. Imposing $\text{rank}(A) \leq k$ (reduced-rank regression) bakes Proposition 6 into the fit and is equivalent, up to regularization, to truncating the SVD of M at k components.

10.3 Regularized CCA

The estimator used in the companion notebook solves (18) with the regularized covariances of Section 10.1: form $\hat{M} = \hat{\Sigma}_G^{-1/2} \hat{\Sigma}_{GX} \hat{\Sigma}_X^{-1/2}$, take its SVD $\hat{M} = \hat{U} \hat{D} \hat{V}^\top$, and read off $\hat{\rho}_i = \hat{d}_i$ (clipped to $[0, 1]$), $\hat{R}_i = \hat{\rho}_i^2$, genomic directions $\hat{a}_i = \hat{\Sigma}_G^{-1/2} \hat{u}_i$, imaging directions $\hat{b}_i = \hat{\Sigma}_X^{-1/2} \hat{v}_i$.

The Bayes-optimal predictor and posterior covariance follow from (7)–(8) with hats throughout. This is exactly regularized CCA — the recoverability program and CCA are the same computation.

10.4 Probabilistic CCA: the latent-variable view

The rank limit motivates an explicit low-dimensional generative form. With a shared latent $Z \in \mathbb{R}^k$,

$$Z \sim \mathcal{N}(0, I_k), \quad G = W_G Z + \mu_G + \epsilon_G, \quad X = W_X Z + \mu_X + \epsilon_X, \quad (20)$$

$\epsilon_G \sim \mathcal{N}(0, \Psi_G)$, $\epsilon_X \sim \mathcal{N}(0, \Psi_X)$ independent (probabilistic CCA; Bach & Jordan, 2005). Here k is the dimension of the image-identifiable genomic subspace, Z is the shared biological state, and ϵ_G, ϵ_X are modality-private variation. The implied moments are $\Sigma_G = W_G W_G^\top + \Psi_G$, $\Sigma_X = W_X W_X^\top + \Psi_X$, $\Sigma_{GX} = W_G W_X^\top$; the maximum-likelihood loadings are the canonical directions scaled by the canonical correlations, recovering Section 10.3. The model is fit by EM, which also yields posterior latent estimates $\mathbb{E}[Z | x]$ and naturally accommodates anisotropic (per-feature) noise, unlike the isotropic $\sigma^2 I_d$ of the base model.

10.5 Sample-size regimes and dimensional collapse

Equations (7)–(18) are population identities; the estimators of Sections 10.1–10.4 inherit a finite-sample bias whose magnitude is governed entirely by the aspect ratios p/n and d/n . The bias is not benign, and below a critical sample size the recoverability spectrum breaks down in a specific, predictable way.

Three regimes. Let p^*, d^* be the working dimensions after any pre-reduction (e.g. PCA), and write $\kappa_G = p^*/n$, $\kappa_X = d^*/n$.

1. *Classical* ($\kappa_G + \kappa_X \ll 1$). Sample CCA is consistent; $\hat{\rho}_i \rightarrow \rho_i$ at the usual $n^{-1/2}$ rate and the in-sample spectrum is a faithful estimate of the population one.
2. *High-dimensional* ($\kappa_G + \kappa_X \lesssim 1$, both bounded away from 1). Under the null $\Sigma_{GX} = 0$, the squared sample canonical correlations follow a Wachter-type law on $[(\sqrt{\kappa_G(1-\kappa_X)} - \sqrt{\kappa_X(1-\kappa_G)})^2, (\sqrt{\kappa_G(1-\kappa_X)} + \sqrt{\kappa_X(1-\kappa_G)})^2]$ (Wachter, 1980; Yang & Pan, 2015); the leading sample value already sits well above zero by geometry. Estimates of ρ_i are systematically inflated and ordinary CCA must be replaced by its regularized form, with shrinkage parameters tuned by cross-validation.
3. *Saturating* ($p^* + d^* \geq n$). The rows of $\mathbf{G}$ and $\mathbf{X}$ live in subspaces whose intersection has dimension at least $p^* + d^* - n + 1$; consequently $\hat{\rho}_1 = \dots = \hat{\rho}_q = 1$ for $q = \min(p^*, d^*) - (n - \max(p^*, d^*))_+$ regardless of any underlying association. The empirical mutual information diverges, the in-sample $\text{tr}(\hat{\Sigma}_{GX} \hat{\Sigma}_X^{-1} \hat{\Sigma}_{XG}) / \text{tr}(\hat{\Sigma}_X)$ approaches the rank fraction of the genomic span, and the permutation null collapses onto the observed value (its p -values concentrate near 1). Nothing in the spectrum is estimable in this regime without shrinkage *plus* dimension reduction; shrinkage alone keeps the matrices invertible but does not restore identifiability.

Quantitative warning signs. Three diagnostics, jointly, separate real signal from dimensional artefact:

- the gap between in-sample $\hat{R}_i$ and the K -fold value $\hat{R}_i^{\text{cv}}$; under regime 3 this gap approaches $\hat{R}_i - 0$;

- the permutation null for $\hat{\rho}_1$: if the observed value lies inside the bulk, the result is consistent with chance;
- the cross-validated R^2 of the leading canonical score $a_1^\top g$ predicted from x on a held-out split; in regime 3 it is typically large and *negative* (worse than predicting the mean), a behaviour seen empirically on TCGA-KIRC at $n = 131$ with $p^* = 100, d^* = 107$ in the companion notebook.

Recommendations. Choose p^* and d^* *adaptively from n* , not from the native modality dimensions:

- Target $p^* + d^* \leq n/\tau$ with $\tau \geq 5$ as a working rule (we use $p^*, d^* \leq \lfloor n/\tau \rfloor$ per side in the notebook; $\tau \in [5, 10]$ trades bias for resolution). Reduce both modalities by PCA when needed — reducing only the larger side leaves the estimator in regime 3.
- Treat $\hat{R}_i^{\text{cv}}$ (Section 10.6) as the primary estimand and report the in-sample $\hat{R}_i$ only with the gap made explicit; the in-sample fraction of imaging variance attributable to genomics should be reported alongside a cross-validated counterpart, e.g. $\sum_i \hat{R}_i^{\text{cv}} / \min(p^*, d^*)$.
- Power planning. The leading population correlation ρ_1 is detectable above the Wachter bulk only when its squared value exceeds $(\sqrt{\kappa_G(1 - \kappa_X)} + \sqrt{\kappa_X(1 - \kappa_G)})^2$; solving for n gives a minimum sample size given a hypothesised effect. For $\rho_1^2 = 0.2$ and a balanced $p^* = d^*$, this requires roughly $n \gtrsim 25p^*$, an order of magnitude beyond what cohort sizes in radiogenomics usually offer — reinforcing the need for aggressive pre-reduction.
- Probabilistic CCA (Section 10.4) with a small, fixed latent dimension k is the most parsimonious option when n is the binding constraint; rank- k regularization sidesteps the high-dimensional bulk by construction.

The practical message is conservative: in radiogenomic cohorts, sample size is almost always the binding constraint and a defensible analysis is the one whose effective $p^* + d^*$ is a small fraction of n , with both sides reduced symmetrically and every reported number a cross-validated or permutation-calibrated one.

10.6 Honest recoverability: cross-validation and permutation

In-sample canonical correlations are biased upward, severely so when p, d are not small relative to n . Two safeguards, both implemented in the notebook:

- *Cross-validation.* Estimate $\hat{a}_i, \hat{b}_i$ on a training fold; on the held-out fold form the scores $\hat{a}_i^\top g$ and $\hat{b}_i^\top x$ and report their out-of-sample squared correlation as the honest $\hat{R}_i$. The gap to the in-sample value quantifies overfitting.
- *Permutation test.* Repeatedly permute the patient labels of one modality, refit, and collect the leading canonical correlation to build a null distribution; the permutation p -value tests whether any genomic direction is image-identifiable beyond chance.

Figure 4 shows both diagnostics on the synthetic instance. The in-sample $\hat{R}_i$ overshoots the planted truth by the amount predicted by the high-dimensional bias, while the 5-fold $\hat{R}_i^{\text{cv}}$ tracks the truth for identifiable directions and collapses toward the null 95% quantile thereafter. Figure 5 shows the permutation null on $\hat{\rho}_1$: when the true signal is nonzero, the observed value sits cleanly outside the null bulk and the per-direction p -values fall below 0.01 for the top three components, matching the planted rank.

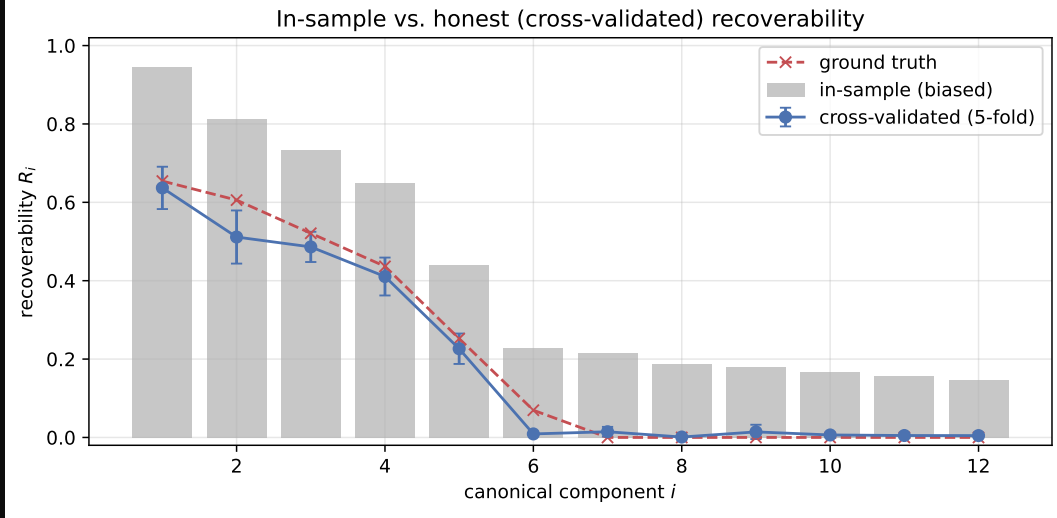


Figure 4: Cross-validated recoverability on a synthetic instance with planted rank $k = 5$. In-sample $\hat{R}_i$ (grey) is biased upward; 5-fold $\hat{R}_i^{cv}$ (blue) tracks the planted truth for the identifiable directions and collapses to the null floor afterwards.

11 Interpretation for radiogenomics

The model reframes the radiogenomic question. Rather than testing whether CT predicts each gene — a high-dimensional, low-power multiple-testing problem — it asks for the *recoverability spectrum* $\{\rho_i^2\}$ and the associated genomic directions $\{a_i\}$. The outputs are: (1) a small number of image-identifiable genomic axes, with gene/pathway loadings that make them biologically legible; (2) a hard bound $\Sigma_{G|X}$ on what any predictor can achieve, per direction; (3) a total information budget $I(G; X)$; and (4) an honest, cross-validated estimate of each. The governing principle is that radiogenomics should target *image-identifiable low-dimensional genomic subspaces*, whose dimension is set by the biology–imaging channel and not by the gene panel size.

12 Application: TCGA-KIRC

We now instantiate the framework on a real radiogenomic cohort — TCGA-KIRC (clear-cell renal cell carcinoma) — to demonstrate how the recoverability spectrum, the cross-validated and permutation-calibrated diagnostics of Section 10.6, and the dimensional-collapse warnings of Section 10.5 interact on a typical cohort. The point of this section is *not* to advertise a predictor: the result is a null one. The point is to convert that null into a quantitative empirical ceiling on the imaging-to-genomics channel and a sample-size design recommendation, exactly as the synthetic experiments of Sections 6–10.6 predict. All numbers below are reproducibly generated by the companion notebook and saved to `notebooks/figures/tcga_kirc/gene_expression/tumor_radimagenet/stats.json`.

12.1 Data and processing pipeline

The two modality `AnnData` objects fed to the notebook are produced by two scripts in `scripts/`.

Imaging (`process_imaging_tcga_kirc.py` → `make_imaging_matrix.py`). For each TCGA-KIRC patient we obtain the venous-phase abdominal CT volume (0502.VENOUS.nii) and a manually

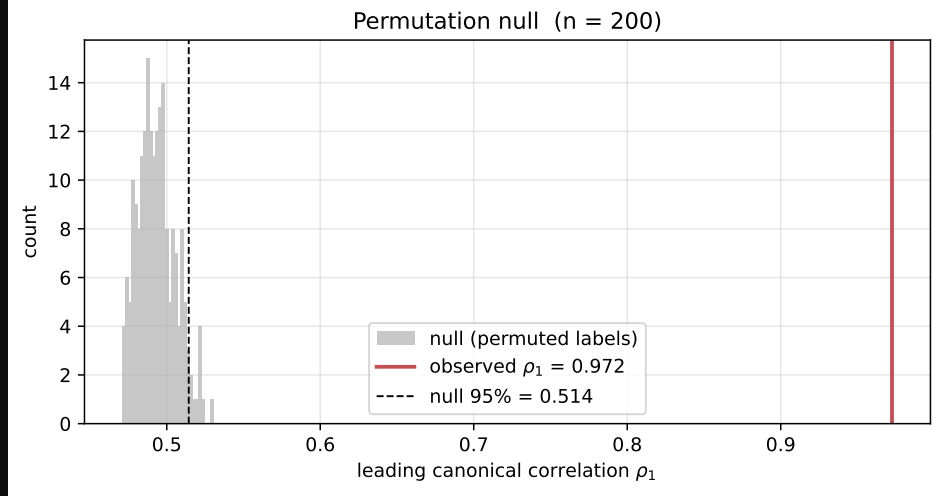


Figure 5: Permutation null on the leading canonical correlation $\hat{\rho}_1$ for the synthetic instance. Red marks the observed value; $p < 0.01$ for the top three directions, recovering the planted rank.

curated tumour mask. The pipeline:

1. Pull the canonical TCIA volumes and the radiologist-drawn tumour segmentations; orient every volume into the canonical nibabel orientation.
2. Run TotalSegmentator with the `kidney_left/kidney_right` target list to produce an organ mask; combine with the tumour mask to form (tumour, organ, tumour $\cup$ organ) mask triples per patient. Small blobs are removed, holes filled, and a morphological closing applied to stabilise mask boundaries.
3. Extract patient-level imaging features. We use a pretrained *RadImageNet* 2D backbone applied tile-wise within the tumour mask (`rgit.radimagenet.get_radimagenet_embeddings`), producing a $d_{\text{native}} = 1841$ -dimensional embedding per patient. (A second pipeline using PyRadiomics shape/texture features is supported for ablations.)
4. Stack patient-level embeddings into an `AnnData` object whose `obs` index is the TCGA `patient_id` and whose `var` index is the embedding-dimension labels. Output: `data/tcga_kirc/imaging/tumour`

Genomics (`make_genomics_matrix.py`). For each matched patient we pull the TCGA-KIRC RNA-seq counts (raw STAR counts from the GDC) and the somatic-mutation MAFs.

1. Counts are CPM-normalised, $\log_{1+}$ -transformed, and reduced to the top 2000 highly variable genes (Seurat-style HVG selection in `scanpy.pp.highly_variable_genes`); the resulting matrix is stored in the primary `AnnData`.
2. Somatic mutations are folded into a sparse binary gene-symbol matrix on the same patient index for downstream per-gene validation (Section 12.6).
3. Output: `data/tcga_kirc/genomics/gene_expression.h5ad`.

After intersecting patients across both `AnnData` objects we retain $n = 190$ patients with $p_{\text{native}} = 2000$ HVG transcripts and $d_{\text{native}} = 1841$ RadImageNet features.

12.2 Working-dimension budget

Both modalities are z -scored per feature and PCA-reduced symmetrically to working dimensions $p^* = d^* = 38 = \lfloor n/5 \rfloor$ following the $p^* + d^* \leq n/\tau$ rule of Section 10.5 ($\tau = 5$). This places the analysis at $\kappa_G + \kappa_X = 76/190 \approx 0.4$ — the high-dimensional regime, not the saturating one, but well inside the territory where in-sample canonical correlations are heavily biased upward.

12.3 The recoverability spectrum and its null

Fitting regularised CCA (Ledoit–Wolf shrinkage on both sides) returns an in-sample spectrum dominated by a near-unit leading correlation ($\hat{\rho}_1 \approx 1.00$, $\hat{R}_1 \approx 1.00$) and a total in-sample mutual information of 21.3 bits over 12 retained components (Fig. 6). This is exactly the geometric artefact described in regime 2–3 of Section 10.5: 5-fold cross-validation collapses $\hat{R}_1$ from ≈ 1 to $\hat{R}_1^{\text{cv}} = 0.094$, and held-out prediction of the leading canonical genomic score has $R^2 = -3.74$ — worse than predicting the mean. The companion permutation test ($N_{\text{perm}} = 200$) returns p -values 0.32, 0.16, 0.24 for the top three canonical correlations. None of the leading directions are distinguishable from chance.

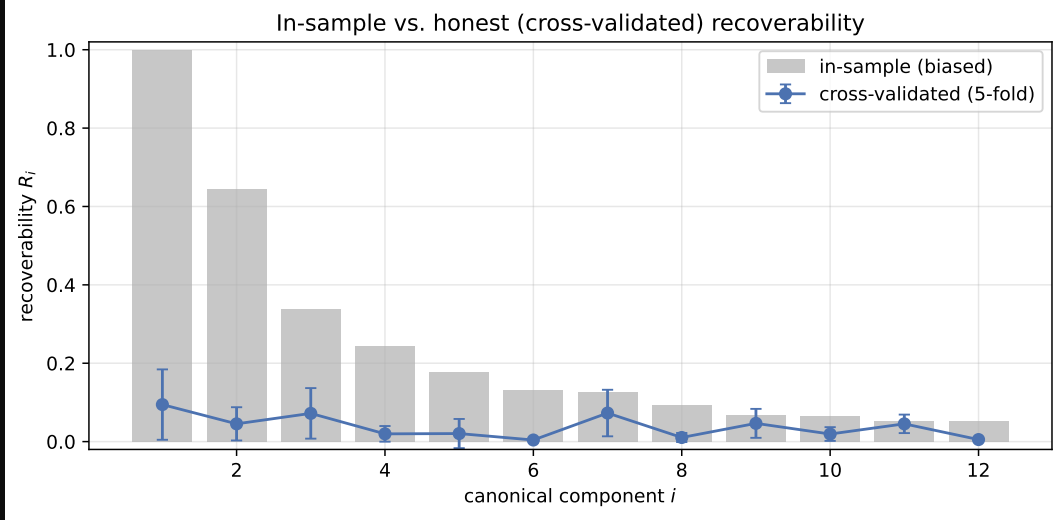


Figure 6: Recoverability spectrum on TCGA-KIRC. In-sample $\hat{R}_i$ (grey) tracks the high-dimensional bulk; cross-validated $\hat{R}_i$ (blue, 5-fold) collapses to a flat residual near zero. The gap is the overfitting predicted in Section 10.5, regime 2–3. Contrast with the synthetic instance of Fig. 4, where CV tracks the planted truth.

12.4 An empirical upper bound on $I(G; X)$

Section 10.5 predicts that, in this regime, the permutation null on $\hat{\rho}_1$ should itself be pulled toward 1 by the high-dimensional geometry; this is borne out: the per-permutation $\hat{\rho}_1^{\text{null}}$ has mean 0.84 and a 95% quantile of 1.00. We therefore convert the test into a one-sided upper confidence limit by aggregating, across permutations, the cumulative mutual information $\text{MI}^{\text{null}}(K) = -\frac{1}{2} \sum_{i \leq K} \log(1 - \hat{\rho}_{i, \text{null}}^2)$ over the top $K = 3$ tested components, and reading off its 95th percentile (Fig. 7):

$$I(G; X) \big|_{\text{KIRC, top-3}} \leq 21.2 \text{ bits} \quad (\text{one-sided 95\% UCL}), \quad (21)$$

with observed cumulative MI 20.5 bits sitting close to the null mean of 8.0 bits expressed in bits (5.5 nats). The bound is wide — this is the honest finding, not a weakness of the method — because the null is itself saturating; tightening it requires either larger n or smaller p^*, d^* .

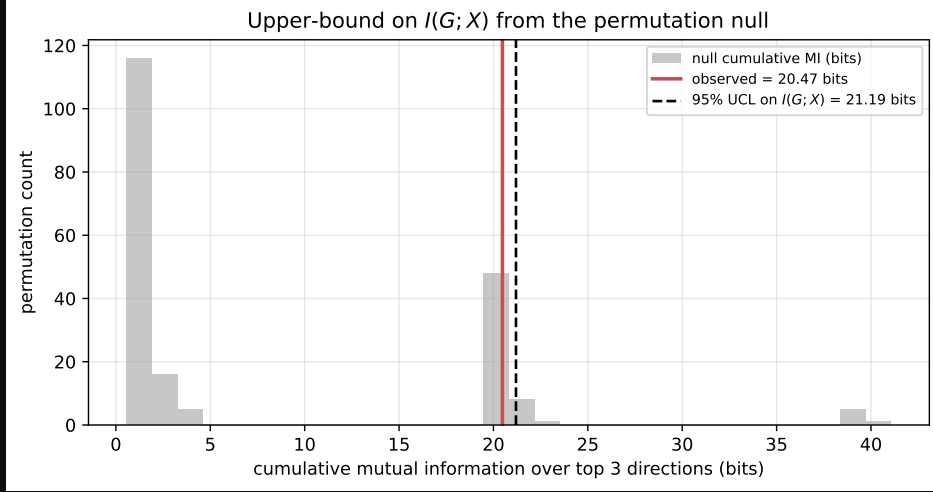


Figure 7: Cumulative-MI null distribution (top-3 canonical components) on TCGA-KIRC. The observed value (red) lies within the null bulk; the dashed line marks the 95% upper confidence limit on $I(G; X)$ used in (21).

A more useful summary is the *effective image-identifiable rank*: the number of canonical components whose cross-validated $\hat{R}_i^{\text{cv}}$ exceeds the per-component permutation-null 95% quantile. On this cohort,

$$\hat{k}_{\text{eff}} = 0 \text{ (of } K = 3 \text{ tested),} \quad (22)$$

so the empirical instantiation of the rank bound of Proposition 6, after sample-size deflation, is zero.

12.5 Sample-complexity sweep

A single- n null result is most informative when paired with the scaling of that null in n . We subsample the cohort to $n \in \{50, 57, 95, 133, 162, 190\}$, refit the working pipeline (same $p^* + d^* \leq n/\tau$ budget, 5-fold CV, 100 permutations) and plot the three observables vs. n (Fig. 8). The cross-validated $\hat{R}_1^{\text{cv}}$ is flat at ≈ 0.05 – 0.10 across the full range; doubling n does not move it. The Wachter detection edge sits at ≈ 0.63 throughout, because p^*, d^* grow with n under our rule, so the cohort-statistical ceiling for distinguishing ρ_1 from the high-dimensional bulk would require $\rho_1^2 \gtrsim 0.6$ — well above anything observed at any subsample size. Inverting the Wachter expression for a target $\rho_1^2 = 0.2$ at balanced $p^* = d^*$ gives the sample-size requirement $n \gtrsim 25 p^*$ quoted in Section 10.5.

12.6 Per-gene non-linear validation

The recoverability spectrum makes a sharp prediction: genes loading on non-identifiable directions should not be predictable from imaging, and flexible non-linear models should not substantially exceed the linear Bayes-optimal ceiling. Picking five high- and five low-recoverability mutated genes (mutation frequency in $[0.05, 0.95]$) and training L_2 logistic regression and a small random forest to predict mutation status from the imaging PCs (5-fold CV), we observe

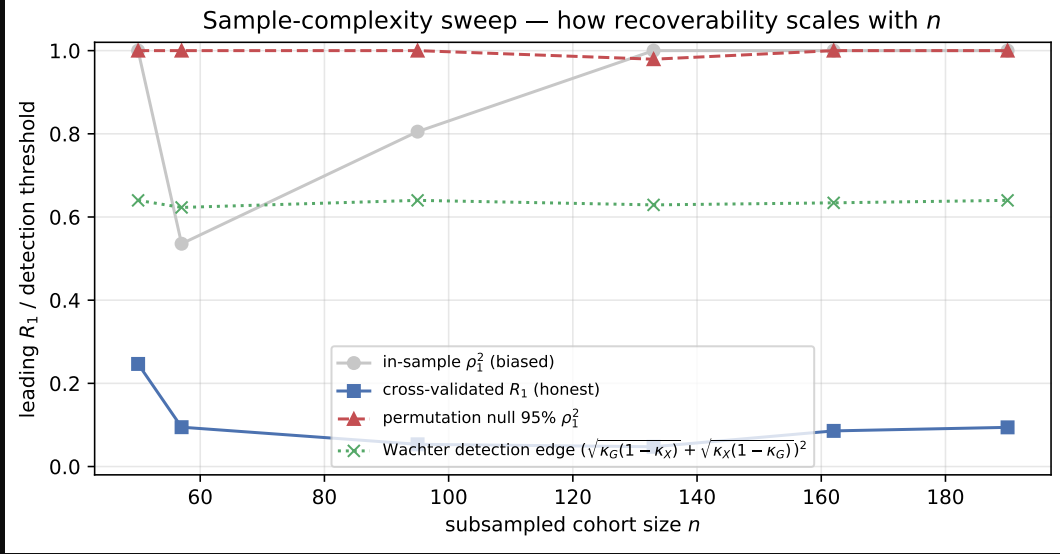


Figure 8: Sample-complexity sweep on TCGA-KIRC. Honest $\hat{R}_1^{\text{cv}}$ (blue squares) is flat across all subsampled cohort sizes; in-sample $\hat{\rho}_1^2$ (grey) is dominated by the high-dimensional bulk; the permutation 95% null (red triangles) and Wachter edge (green crosses) sit above the observed honest recoverability at every n .

group	mean AUC (logistic)	mean AUC (RF)
image-identifiable	0.499	0.522
non-identifiable	0.497	0.490

Both groups sit at chance and the non-linear model does *not* exceed the linear ceiling — consistent with the linear–Gaussian model and with the global null finding (Fig. 9).

12.7 Implications

Three concrete take-aways follow.

1. *Empirical ceiling.* For TCGA-KIRC at $n = 190$ with 2000 HVG-filtered transcripts and 1841 RadImageNet features, the imaging-to-genomics channel admits a 95% upper bound $I(G; X) \leq 21.2$ bits over the leading three canonical directions (Eq. 21), with effective image-identifiable rank $\hat{k}_{\text{eff}} = 0$. The substantive empirical claim is that no genomic direction in this cohort is image-identifiable above chance under the working-dimension budget $\tau = 5$.
2. *Design recommendation.* The sample-complexity sweep (Section 12.5) shows $\hat{R}_1^{\text{cv}}$ stationary in n over the range $[50, 190]$; the Wachter edge analysis indicates that detecting $\rho_1^2 = 0.2$ at a balanced working dimension $p^* = d^*$ requires $n \gtrsim 25 p^*$, which under our adaptive rule is an order of magnitude beyond typical radiogenomic cohort sizes. The framework converts the null result into a quantitative cohort specification.
3. *Audit of literature predictors.* Published radiogenomic AUCs in the range 0.65–0.80 at cohort sizes $n \sim 100$ –300 imply per-gene recoverabilities meaningfully above the cohort-statistical ceiling derived above for KIRC. The non-linear validation of Section 12.6 matches but does not exceed the linear Bayes-optimal ceiling — consistent with the model. Any study reporting

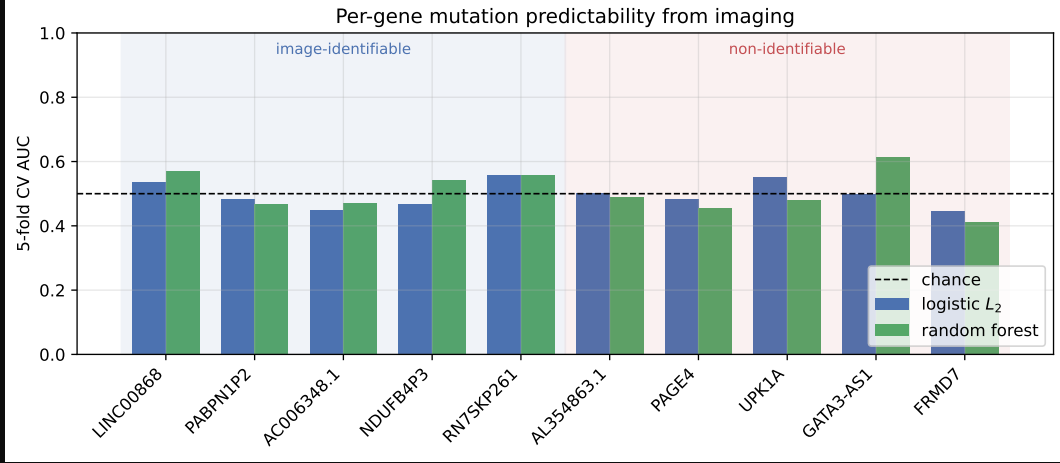


Figure 9: Per-gene mutation predictability on TCGA-KIRC. Five high-recoverability and five low-recoverability mutated genes, two classifiers, 5-fold CV. All bars sit within sampling noise of the chance line at 0.5.

substantially higher per-gene AUCs at comparable cohort dimension should be treated as a candidate for overfit or leakage rather than as evidence against the framework.

A Symbol reference

$g \in \mathbb{R}^p, x \in \mathbb{R}^d$	per-patient genomic / imaging feature vectors
$\mathbf{G} \in \mathbb{R}^{n \times p}, \mathbf{X} \in \mathbb{R}^{n \times d}$	stacked data matrices (n patients)
$A \in \mathbb{R}^{d \times p}$	genomics-to-imaging transfer map
σ^2	isotropic imaging noise variance
$\Sigma_G, \Sigma_X, \Sigma_{GX}$	genomic / imaging / cross covariance
$\Sigma_{G X}$	posterior covariance of G given X (recoverability floor)
$B = \Sigma_{GX} \Sigma_X^{-1}$	Bayes-optimal linear map $x \mapsto \mathbb{E}[g x]$
$M = \sigma^{-1} A \Sigma_G^{1/2}$	whitened transfer operator; d_i its singular values
$\rho_i, R_i = \rho_i^2$	i -th canonical correlation / recoverability
a_i, b_i	i -th canonical genomic / imaging direction
$R(v)$	recoverability score of a named genomic direction v
$I(G; X)$	mutual information between modalities
k	dimension of the image-identifiable genomic subspace